

Hints & Solutions

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Q1 (3) - Order, Degree and Formation of DE

We have,

$$\sqrt[3]{\frac{dy}{dx}} - 4\frac{d^2y}{dx^2} - 7x = 0$$

$$\Rightarrow \sqrt[3]{\frac{dy}{dx}} = 4\frac{d^2y}{dx^2} + 7x$$

$$\frac{dy}{dx} = \left(4\frac{d^2y}{dx^2} + 7x\right)^3$$

Hence, order $a = 2$ and degree $b = 3$

$$\therefore a + b = 2 + 3 = 5$$

Q2 (3) - Order, Degree and Formation of DE

We have, $y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{x+c_5}$

$$y = (c_1 + c_2) \cos(x + c_3) - (c_4 e^{c_5} \cdot e^x)$$

$$y = A \cos(x + c_3) - B e^x \quad \left[\begin{array}{l} \because c_1 + c_2 = A \\ \text{and } c_4 e^{c_5} = B \end{array} \right]$$

The above equation have three arbitrary constants A, B and c_3 .

Hence, the order of differential equation is 3.

Q3 (2) - Order, Degree and Formation of DE

We have, $y = a + bx + ce^{-x}$

$$\therefore \frac{dy}{dx} = b - ce^{-x}$$

$$\Rightarrow \frac{d^2y}{dx^2} = ce^{-x}$$

$$\Rightarrow \frac{d^3y}{dx^3} = -ce^{-x}$$

$$\text{Now, } \frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} = 0$$

$$\Rightarrow y''' + y'' = 0$$

Q4 (3) - Order, Degree and Formation of DE

The equation of family of circle with the fixed radius 5 units and centre on the line $y = 2$, is

$$(x - a)^2 + (y - 2)^2 = 25$$

On differentiating Eq.(i) w.r.t. x , we get

$$2(x - a) + 2(y - 2)y' = 0$$

$$\Rightarrow (x - a) = -(y - 2)y'$$

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On putting the value of $(x - a)$ in Eq. (i), we get

$$(y - 2)^2 y'^2 + (y - 2)^2 = 25$$

$$\Rightarrow (y - 2)^2 y'^2 = 25 - (y - 2)^2$$

Q5 (4) - Variable Separable Form and Reducible to Variable Separable Form

Given, $xy \frac{dy}{dx} = \frac{1+y^2}{1+x^2} (1+x+x^2)$

$$\Rightarrow \frac{y}{1+y^2} dy = \frac{1+x+x^2}{x(1+x^2)}$$

$$\Rightarrow \frac{y}{1+y^2} dy = \left(\frac{1}{x} + \frac{1}{1+x^2} \right) dx$$

On integrating both sides, we get

$$\int \frac{y}{1+y^2} dy = \int \left(\frac{1}{x} + \frac{1}{1+x^2} \right) dx$$

$$\Rightarrow \frac{1}{2} \log(1+y^2) = \log x + \tan^{-1} x + C$$

$$\Rightarrow \log \sqrt{1+y^2} - \log x = \tan^{-1} x + C$$

$$\Rightarrow \log \frac{\sqrt{1+y^2}}{x} = \tan^{-1} x + C$$

$$\Rightarrow \frac{\sqrt{1+y^2}}{x} = C e^{\tan^{-1} x}$$

$$\Rightarrow \sqrt{1+y^2} = C x e^{\tan^{-1} x}$$

Q6 (2) - Variable Separable Form and Reducible to Variable Separable Form

We have,

$$\sin^{-1} \left(\frac{dy}{dx} \right) = x + y$$

$$\Rightarrow \frac{dy}{dx} = \sin(x + y)$$

Put $x + y = v \Rightarrow 1 + \frac{dy}{dx} = \frac{dv}{dx}$

$$\therefore \frac{dv}{dx} - 1 = \sin v$$

$$\Rightarrow \frac{dv}{1 + \sin v} = dx$$

On integrating both sides, we get

$$\int \frac{1 - \sin v}{\cos^2 v} dv = \int dx$$

$$\Rightarrow \int (\sec^2 v - \sec v \tan v) dv = \int dx$$

$$\Rightarrow \tan v - \sec v = x + C$$

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$$\Rightarrow \tan(x+y) - \sec(x+y) = x + C$$

$$\Rightarrow x = \tan(x+y) - \sec(x+y) + C$$

Q7 (1) - Variable Separable Form and Reducible to Variable Separable Form

$$\text{Given, } y \log y + xy' = 0$$

$$\Rightarrow y \log y + x \frac{dy}{dx} = 0$$

On integrating both sides, we get

$$\int \frac{dy}{y \log y} = - \int \frac{dx}{x}$$

$$\Rightarrow \log(\log y) = -\log x + C$$

$$\Rightarrow \log \log y + \log x = C$$

$$\Rightarrow x(\log y) = C$$

$$\text{It is given that, } y(1) = e$$

$$\therefore 1 \cdot \log e = C \Rightarrow C = 1$$

$$\text{Hence, } x(\log y) = 1$$

Q8 (1) - Variable Separable Form and Reducible to Variable Separable Form

$$\text{We have, } \frac{2+\sin x}{y+1} \cdot \frac{dy}{dx} = -\cos x$$

$$\Rightarrow \int \frac{dy}{y+1} = - \int \frac{\cos x}{2+\sin x} dx$$

$$\Rightarrow \log(y+1) = -\log(2+\sin x) + \log C$$

$$\Rightarrow \log(y+1) + \log(2+\sin x) = \log C$$

$$\Rightarrow (y+1)(2+\sin x) = C$$

$$\text{It is given that, } y(0) = 1$$

Put $x = 0$ and $y = 1$, then

$$2(2+\sin 0) = C \Rightarrow C = 4$$

So, the equation is

$$(y+1)(2+\sin x) = 4$$

Now, put $x = \pi/2$

$$\therefore (y+1) \left(2 + \sin \frac{\pi}{2} \right) = 4$$

$$\Rightarrow (y+1)(3) = 4 \Rightarrow y = \frac{1}{3}$$

Q9 (3) - Variable Separable Form and Reducible to Variable Separable Form

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Given, $\frac{dP}{dx} = 100 - 12\sqrt{x}$

$$\Rightarrow dP = (100 - 12\sqrt{x})dx$$

On integrating both sides, we get

$$\int dP = \int (100 - 12\sqrt{x})dx$$

$$\Rightarrow P = 100x - 8(x)^{3/2} + C$$

When $x = 0, P = 2000$, then

$$C = 2000$$

Now, when $x = 25$, then

$$\begin{aligned} P &= 100(25) - 8(25)^{3/2} + 2000 \\ &= 2500 - 8(125) + 2000 = 3500 \end{aligned}$$

Q10 (4) - Variable Separable Form and Reducible to Variable Separable Form

We have, $f'(x) < 2f(x) \Rightarrow \frac{f'(x)}{f(x)} < 2$

On integrating both sides, we get

On integrating both sides, we get

$$\log f(x) < 2x + C$$

$$\Rightarrow f(x) < e^{2x+C}$$

$$\Rightarrow f(x) < e^{2x-1} \left[\because f\left(\frac{1}{2}\right) = 1 \right]$$

Hence, $f(x) > 0$

$$\therefore 0 < f(x) < e^{2x-1}$$

$$0 < \int_{1/2}^1 f(x)dx < \int_{1/2}^1 e^{2x-1}dx$$

$$\Rightarrow 0 < \int_{1/2}^1 f(x)dx < \left[\frac{e^{2x-1}}{2} \right]_{1/2}^1$$

$$0 < \int_{1/2}^1 f(x)dx < \frac{e-1}{2}$$

Q11 (3) - Homogeneous Form and Reducible to Homogeneous Form

Given, $x \frac{dy}{dx} = y(\log y - \log x + 1)$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} \left(\log \frac{y}{x} + 1 \right)$$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$

$$\therefore v + x \frac{dv}{dx} = v(\log v + 1) \Rightarrow x \frac{dv}{dx} = v \log v$$

$$\Rightarrow \frac{dv}{v \log v} = \frac{dx}{x} \Rightarrow \int \frac{dv}{v \log v} = \int \frac{dx}{x}$$

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$$\Rightarrow \log(\log v) = \log x + \log C$$

$$\Rightarrow \log v = Cx$$

$$\Rightarrow \log\left(\frac{y}{x}\right) = Cx \quad [\because v = y/x]$$

Q12 (1) - Homogeneous Form and Reducible to Homogeneous Form

Slope of the curve at point $P(x, y)$ is $\frac{y}{x} + \sec\left(\frac{y}{x}\right)$.

$$\therefore \frac{dy}{dx} = \frac{y}{x} + \sec\left(\frac{y}{x}\right)$$

$$\Rightarrow v + x \frac{dv}{dx} = v + \sec v \quad \left[\because y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx} \right]$$

$$x \frac{dv}{dx} = \sec v$$

On integrating both sides, we get

$$\Rightarrow \int \frac{dv}{\sec v} = \int \frac{dx}{x} \Rightarrow \int \cos v dv = \int \frac{dx}{x}$$

$$\Rightarrow \sin v = \log x + C$$

$$\Rightarrow \sin y/x = \log x + C$$

When $x = 1, y = \pi/6$, then $C = \frac{1}{2}$

$\therefore$ Equation of the curve is

$$\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$$

Q13 (1) - Homogeneous Form and Reducible to Homogeneous Form

Given, $(3xy + y^2) dx + (x^2 + xy) dy = 0$

$$\Rightarrow \frac{dy}{dx} = -\frac{(3xy+y^2)}{x^2+xy}$$

$$\Rightarrow v + x \frac{dv}{dx} = -\frac{(3v+v^2)}{1+v} \quad \left[\because y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx} \right]$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-3v-v^2}{1+v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-3v-v^2-v-v^2}{1+v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-4v-2v^2}{1+v}$$

$$\Rightarrow \frac{1+v}{2v+v^2} dv = -\frac{2dx}{x}$$

On integrating both sides, we get

$$\int \frac{1+v}{v(2+v)} dv = -2 \int \frac{dx}{x}$$

$$\Rightarrow \int \frac{v+1}{v^2+2v} dv = -2 \int \frac{dx}{x}$$

$$\Rightarrow \frac{1}{2} \log(v^2 + 2v) = -2 \log x + \log C$$

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$$\Rightarrow \frac{1}{2} \log \left(\frac{y^2}{x^2} + \frac{2y}{x} \right) + 2 \log x = \log C$$

$$\Rightarrow \log \left[\frac{(y^2 + 2xy)}{x^2} \times x^4 \right] = 2 \log C$$

$$\Rightarrow x^2 (y^2 + 2xy) = C$$

Q14 (1) - Homogeneous Form and Reducible to Homogeneous Form

We have,

$$xy \cdot \log \left(\frac{x}{y} \right) dx + \left\{ y^2 - x^2 \log \left(\frac{x}{y} \right) \right\} dy = 0$$

$$\Rightarrow \frac{dx}{dy} = \frac{-\{y^2 - x^2 \log(x/y)\}}{xy \log(x/y)}$$

$$\Rightarrow v + y \frac{dv}{dy} = -\frac{(1 - v^2 \log v)}{v \log v} \left[\because x = vy \Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy} \right]$$

$$\Rightarrow y \frac{dv}{dy} = \frac{-1 + v^2 \log v - v^2 \log v}{v \log v}$$

$$\Rightarrow v \log v dv = -\frac{dy}{y}$$

On integrating both sides, we get

$$\int v \log v dv = \int -\frac{dy}{y}$$

$$\Rightarrow \log v \int v dv - \int \frac{d \log v}{dv} \cdot \int v dv \cdot dv = -\log y + \log C$$

$$\Rightarrow \frac{v^2}{2} \log v - \int \frac{1}{v} \cdot \frac{v^2}{2} dv = -\log y + \log C$$

$$\Rightarrow \frac{v^2}{2} \log v - \frac{v^2}{4} = -\log y + \log C$$

$$\Rightarrow 2v^2 \log v - v^2 = 4 \log C/x$$

$$\Rightarrow 2x^2 \log \left(\frac{x}{y} \right) - x^2 = 4y^2 \log \left(\frac{C}{x} \right)$$

Q15 (4) - Homogeneous Form and Reducible to Homogeneous Form

$$\text{We have. } \frac{dy}{dx} = \frac{2x-y+3}{x+2y+4} \dots (i)$$

The given equation is non-homogeneous.

On solving $2x - y + 3 = 0$ and $x + 2y + 4 = 0$, we get

$$x = -2 \text{ and } y = -1$$

$$\text{Put } x = -2 + u \text{ and } y = -1 + v \Rightarrow dx = du \text{ and } dy = dv$$

$$\Rightarrow \frac{dy}{dx} = \frac{dv}{du}$$

Eq. (i) reduces to

$$\frac{dv}{du} = \frac{2(-2+u) - (-1+v) + 3}{(-2+u) + 2(-1+v) + 4}$$

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$$\Rightarrow \frac{dv}{du} = \frac{2u-v}{u+2v}$$

$$\Rightarrow t + u \frac{dt}{du} = \frac{2-t}{1+2t} \left[\because v = ut \Rightarrow \frac{dv}{du} = t + u \frac{dt}{du} \right]$$

$$\Rightarrow u \frac{dt}{du} = \frac{2-t-t-2t^2}{1+2t}$$

On integrating both sides, we get

$$\frac{1+2t}{t^2+t+1} dt = -2 \frac{du}{u} \Rightarrow \int \frac{2t+1}{t^2+t+1} dt = -2 \int \frac{du}{u}$$

$$\Rightarrow \log(t^2 + t - 1) = \log \frac{C}{u^2} \Rightarrow t^2 + t - 1 = \frac{C}{u^2}$$

$$\Rightarrow \frac{v^2}{u^2} + \frac{v}{u} - 1 = \frac{C}{u^2} \left[\because t = \frac{v}{u} \right]$$

$$\Rightarrow v^2 + uv - u^2 = C$$

$$\Rightarrow (y+1)^2 + (x+2)(y+1) - (x+2)^2 = C$$

$$\Rightarrow y^2 + 2y + 1 + xy + x + 2y + 2 - x^2 - 4x - 4 = C$$

$$\Rightarrow y^2 - x^2 + xy + 4y - 3x - 1 = C$$

Q16 (3) - Homogeneous Form and Reducible to Homogeneous Form

$$\text{Given, } (2x^2 + 3y^2 - 7)x \cdot dx - (3x^2 + 2y^2 + 8)y \cdot dy = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{(2x^2+3y^2-7)x}{(3x^2+2y^2-8)y}$$

On solving $2x^2 + 3y^2 - 7 = 0$ and $3x^2 + 2y^2 - 8 = 0$, we get

$$x^2 = 2 \text{ and } y^2 = 1$$

Now, put $x^2 = u + 2$ and $y^2 = v + 1$

$$\Rightarrow 2x dx = du \text{ and } 2y dy = dv$$

$$\Rightarrow \frac{y dy}{x dx} = \frac{dv}{du}$$

$$\text{Eq. (i) reduces to } \frac{dv}{du} = \frac{2u+3v}{3u+2v}$$

$$\Rightarrow t + u \frac{dt}{du} = \frac{2+3t}{3+2t} \left[\because v = tu \Rightarrow \frac{dv}{du} = t + u \frac{dt}{du} \right]$$

$$\Rightarrow u \frac{dt}{du} = \frac{2+3t-3t-2t^2}{3+2t}$$

$$\frac{3+2t}{1-t^2} dt = \frac{2du}{v}$$

On integrating both sides, we get

$$\int \left(\frac{5/2}{1-t} + \frac{1/2}{1+t} \right) dt = 2 \int \frac{du}{u}$$

$$\Rightarrow -\frac{5}{2} \log|1-t| + \frac{1}{2} \log|1+t| = 2 \log u + \log C$$

$$\Rightarrow \frac{1}{2} \log \frac{1+t}{(1-t)^5} = \log u^2 C$$

$$\Rightarrow \log \frac{1+t}{(1-t)^5} = \log u^4 C$$

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$$\Rightarrow \frac{1+t}{(1-t)^5} = u^4 C$$

$$\Rightarrow \frac{u+v}{(u-v)^5} = C \left[\because t = \frac{v}{u} \right]$$

$$\Rightarrow u + v = C(u - v)^5$$

$$\Rightarrow x^2 + y^2 - 3 = C(x^2 - y^2 - 1)^5$$

$$\Rightarrow \left[\because u = x^2 - 2, v = y^2 - 1 \right]$$

Q17 (1) - Homogeneous Form and Reducible to Homogeneous Form

We have,

$$\frac{dy}{dx} + \frac{\cos x(3 \cos y - 7 \sin x - 3)}{\sin y(3 \sin x - 7 \cos y + 7)} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\cos x(3 \cos y - 7 \sin x - 3)}{\sin y(3 \sin x - 7 \cos y + 7)}$$

On solving $3 \cos y - 7 \sin x - 3 = 0$ and

$3 \sin x - 7 \cos y + 7 = 0$, we get

$$\sin x = 0, \cos y = 1$$

Now, put $\sin x = u$ and $\cos y = v + 1$

$$\Rightarrow \cos x dx = du \text{ and } -\sin y dy = dv$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\cos x}{\sin y} \frac{dv}{du}$$

$$\text{Eq. (i) reduces to } \frac{dv}{du} = \frac{3v-7u}{3u-7v}$$

$$\Rightarrow t + u \frac{dt}{du} = \frac{3t-7}{3-7t} \left[\because v = ut \Rightarrow \frac{dv}{du} = t + u \frac{dt}{du} \right]$$

$$\Rightarrow u \frac{dt}{du} = \frac{3t-7-3t+7t^2}{3-7t}$$

$$\Rightarrow \frac{3-7t}{t^2-1} dt = 7 \frac{du}{u}$$

On integrating both sides, we get

$$\int \left(\frac{-5}{t+1} - \frac{2}{t-1} \right) dt = 7 \int \frac{du}{u}$$

$$\Rightarrow -5 \log |t+1| - 2 \log |t-1| = 7 \log u + C$$

$$\Rightarrow 5 \log |t+1| + 2 \log |t-1| = -7 \log u + C$$

$$\Rightarrow \log(t+1)^5 (t-1)^2 = \log C/u^7$$

$$\Rightarrow (t+1)^5 (t-1)^2 = \frac{C}{u^7}$$

$$\Rightarrow \frac{(u+v)^5 (v-u)^2}{u^7} = \frac{C}{u^7} \left[\because t = \frac{v}{u} \right]$$

$$\Rightarrow (u+v)^5 (v-u)^2 = C$$

$$\Rightarrow (\sin x + \cos y - 1)^5 (\cos y - \sin x - 1)^2 = C$$

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Q18 (4) - Linear Differential Equations and Bernoulli's Equation

We have, $(1+t) \frac{dy}{dt} - ty = 1$

$$\Rightarrow \frac{dy}{dt} - \left(\frac{t}{1+t}\right)y = \frac{1}{1+t}$$

which is a linear differential equation of the form

$$\frac{dy}{dt} + Py = Q(t)$$

where, $P = \frac{-t}{1+t}$ and $Q = \frac{1}{1+t}$

$$\text{Now, } \int P dt = \int \frac{-t}{1+t} dt = - \int \frac{t+1}{t+1} dt + \int \frac{1}{1+t} dt$$

$$= -t + \log(1+t)$$

$$\therefore \text{IF} = e^{\int P dt} = e^{-t + \log(1+t)} = e^{-t} \cdot e^{\log(1+t)} = e^{-t}(1+t)$$

So, the general solution is

$$y \cdot \text{IF} = \int Q \cdot \text{IF} dt$$

$$\Rightarrow ye^{-t}(1+t) = \int \frac{1}{1+t} \cdot e^{-t}(1+t) dt$$

$$\Rightarrow ye^{-t}(1+t) = -e^{-t} + C$$

When $t = 0, y = -1$, then $C = 0$

$$\therefore y = \frac{-1}{1+t}$$

$$\text{When } t = 1, \text{ then } y = \frac{-1}{2}$$

Q19 (1) - Linear Differential Equations and Bernoulli's Equation

It is given that, $\frac{d(p(x))}{dx} = p(x)$ for all $x \geq 1$

$$\Rightarrow \frac{dp(x)}{dx} - p(x) > 0$$

On multiplying both sides by e^{-x} , we get

$$e^{-x} \frac{dp(x)}{dx} - e^{-x} p(x) > 0$$

$$\Rightarrow d[p(x)e^{-x}] > 0$$

Hence, $p(x)e^{-x}$ is increasing, $\forall x \in [1, \infty]$.

$$\text{Now, } p(1) = 0$$

$$\therefore p(x) > 0 \text{ for all } x > 1$$

Q20 (1) - Linear Differential Equations and Bernoulli's Equation

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We have, $x^2 \frac{dy}{dx} \cos\left(\frac{1}{x}\right) - y \sin\left(\frac{1}{x}\right) = -1$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x^2} \tan\left(\frac{1}{x}\right) = \frac{-\sec(1/x)}{x^2}$$

which is a linear differential equation.

where, $P = \frac{-\tan(1/x)}{x^2}$ and $Q = \frac{-\sec(1/x)}{x^2}$

Now, IF = $e^{\int \frac{-\tan(1/x)}{x^2} dx} = e^{\log \sec\left(\frac{1}{x}\right)} = \sec\left(\frac{1}{x}\right)$

So, the solution is

$$y \cdot \sec \frac{1}{x} = \int \frac{-\sec^2\left(\frac{1}{x}\right)}{x^2} dx$$

$$\Rightarrow y \sec \frac{1}{x} = \tan \frac{1}{x} + C$$

$$\Rightarrow y = \sin\left(\frac{1}{x}\right) + C \cos\left(\frac{1}{x}\right)$$

When $x \rightarrow \infty$ and $y \rightarrow -1$, then $C = -1$

Hence, the required solution is

$$y = \sin \frac{1}{x} - \cos \frac{1}{x}$$

Q21 (3) - Linear Differential Equations and Bernoulli's Equation

Given, $1 + y^2 + (x - e^{\tan^{-1}y}) \frac{dy}{dx} = 0$

$$\therefore (x - e^{\tan^{-1}y}) \frac{dy}{dx} = -(1 + y^2)$$

$$\Rightarrow \frac{dx}{dy} = \frac{-(x - e^{\tan^{-1}y})}{1 + y^2}$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{1 + y^2} = \frac{e^{\tan^{-1}y}}{1 + y^2}$$

Now, IF = $e^{\int \frac{1}{1 + y^2} dy} = e^{\tan^{-1}y}$

Hence, the solution of given differential equation is

$$xe^{\tan^{-1}y} = \int \frac{e^{2 \tan^{-1}y}}{1 + y^2} dy + C$$

$$\Rightarrow xe^{\tan^{-1}y} = \frac{1}{2} e^{2 \tan^{-1}y} + C$$

$$\Rightarrow 2xe^{\tan^{-1}y} = e^{2 \tan^{-1}y} + C$$

Q22 (3) - Linear Differential Equations and Bernoulli's Equation

We have, $(2x - 10y^3) \frac{dy}{dx} + y = 0$

$$\Rightarrow \frac{dy}{dx} = \frac{-y}{2x - 10y^3}$$

$$\Rightarrow \frac{dx}{dy} = \frac{10y^3 - 2x}{y}$$

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$$\Rightarrow \frac{dx}{dy} + \frac{2x}{y} = 10y^2$$

$$\text{IF} = e^{\int \frac{2}{y} dy} = e^{2 \log y} = y^2$$

Hence, the solution of given differential equation is

$$xy^2 = \int 10y^4 dy + C$$

$$\Rightarrow xy^2 = 2y^5 + C$$

Q23 (4) - Linear Differential Equations and Bernoulli's Equation

$$\text{We have, } y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{y^2} = \frac{1}{y^3}$$

$$\therefore \text{IF} = e^{\int \frac{1}{y^2} dy} = e^{-\frac{1}{y}}$$

Hence, the solution of given differential equation is

$$xe^{-1/y} = \int \frac{e^{-1/y}}{y^3} dy$$

$$\Rightarrow xe^{-1/y} = -\int e^t \cdot t dt \left[\because \text{put } \frac{-1}{y} = t \Rightarrow \frac{1}{y^2} dy = dt \right]$$

$$\Rightarrow xe^{-1/y} = -e^t(t-1) + C$$

$$\Rightarrow xe^{-1/y} = e^{-1/y} + \frac{e^{-1/y}}{y} + C$$

$$\text{When } x = 1 \text{ and } y = 1, \text{ then } C = \frac{-1}{e}$$

$$\therefore x = 1 + \frac{1}{y} - \frac{e^{1/y}}{e}$$

Q24 (1) - Linear Differential Equations and Bernoulli's Equation

$$\text{We have, } \lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$$

$$\Rightarrow \lim_{t \rightarrow x} \frac{2tf(x) - x^2 f'(t)}{1} = 1 \text{ [applying L' Hospital's rule]}$$

$$\Rightarrow 2xf(x) - x^2 f'(x) = 1$$

$$\Rightarrow x^2 f'(x) = 2xf(x) - 1$$

$$\Rightarrow f'(x) - \frac{2f(x)}{x} = \frac{-1}{x^2}$$

which is a form of linear differential equation. where, $P = \frac{-2}{x}$ and $Q = \frac{-1}{x^2}$

$$\therefore \text{IF} = e^{\int -2/x dx} = e^{-2 \log x} = \frac{1}{x^2}$$

Hence, the solution of given differential equation is

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$$f(x) \cdot \frac{1}{x^2} = \int \frac{-1}{x^4} dx$$

$$\Rightarrow f(x) \cdot \frac{1}{x^2} = \frac{1}{3x^3} + C$$

$$\Rightarrow f(x) = \frac{1}{3x} + Cx^2$$

$$\Rightarrow f(1) = \frac{1}{3} + C = 1 \Rightarrow C = \frac{2}{3}$$

$$\therefore f(x) = \frac{1}{3x} + \frac{2x^2}{3}$$

Q25 (1) - Linear Differential Equations and Bernoulli's Equation

We have, $y' - y \tan x = 2x \sec x$

$$\Rightarrow \frac{dy}{dx} - y \tan x = 2x \sec x$$

which is a form of linear differential equation.

where, $P = -\tan x$ and $Q = 2 \times \sec x$

$$\therefore IF = e^{\int -\tan x dx} = e^{\log \cos x} = \cos x$$

So, the solution of given differential equation is

$$y \cos x = \int 2x \sec x \cos x dx + C$$

$$y \cos x = x^2 + C$$

When $x = 0$ and $y = 0$, then $C = 0$

$$\therefore y = x^2 \sec x$$

$$\Rightarrow y\left(\frac{\pi}{4}\right) = \left(\frac{\pi}{4}\right)^2 \sec \frac{\pi}{4} = \frac{\pi^2}{16} \times \sqrt{2} = \frac{\pi^2}{8\sqrt{2}}$$

Q26 (3) - Exact Form

We have,

$$x dy + y dx + 2x^3 dx = 0$$

$$\Rightarrow d(xy) + 2x^3 dx = 0$$

On integrating both sides, we get

$$\int d(xy) + \int 2x^3 dx = 0$$

$$\therefore xy + \frac{x^4}{2} = C$$

Q27 (1) - Exact Form

Hints & Solutions

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Given, $x dx + y dy = a (x^2 + y^2) dy$

$$\Rightarrow \frac{x dx + y dy}{x^2 + y^2} = a dy$$

$$\Rightarrow \frac{1}{2} d \log(x^2 + y^2) = a dy$$

On integrating both sides, we get

$$\int d [\log(x^2 + y^2)] = 2 \int a dy$$

$$\therefore \log(x^2 + y^2) = 2ay + C$$

Q28 (1) - Exact Form

We have, $y dx - x dy + (1 + x^2) dx + x^2 \sin y dy = 0$

On dividing by x^2 , we get

$$\frac{y dx - x dy}{x^2} + \left(\frac{1}{x^2} + 1\right) dx + \sin y dy = 0$$

$$- \left(\frac{x dy - y dx}{x^2}\right) + \left(\frac{1}{x^2} + 1\right) dx + \sin y dy = 0$$

On integrating both sides, we get

$$- \int d\left(\frac{y}{x}\right) + \int \left(\frac{1}{x^2} + 1\right) dx + \int \sin y dy = C$$

$$\Rightarrow -\frac{y}{x} + x - \frac{1}{x} - \cos y = C$$

$$\therefore \frac{y}{x} + \frac{1}{x} - x + \cos y + C = 0$$

Q29 (2) - Exact Form

Given,

$$\left(1 + x\sqrt{x^2 + y^2}\right) dx + \left(-1 + \sqrt{x^2 + y^2}\right) y dy = 0$$

$$dx + x\sqrt{x^2 + y^2} dx - y dy + y\sqrt{x^2 + y^2} dy = 0$$

$$\Rightarrow dx - y dy + x\sqrt{x^2 + y^2} dx + y\sqrt{x^2 + y^2} dy = 0$$

$$dx - y dy + \frac{1}{3} d(x^2 + y^2)^{3/2} = 0$$

On integrating both sides, we get

$$\int dx - \int y dy + \int \frac{1}{3} d(x^2 + y^2)^{3/2} = 0$$

$$\Rightarrow x - \frac{y^2}{2} + \frac{1}{3} (x^2 + y^2)^{3/2} = C$$

$$\therefore 2x - y^2 + \frac{2}{3} (x^2 + y^2)^{3/2} = C$$

Q30 (1) - Exact Form

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$$\begin{aligned} \text{We have, } \frac{xdy}{x^2+y^2} &= \left(\frac{y}{x^2+y^2} - 1 \right) dx \\ \Rightarrow \frac{xdy}{x^2+y^2} - \frac{ydx}{x^2+y^2} &= -dx \\ \Rightarrow \frac{xdy-ydx}{x^2+y^2} &= -dx \Rightarrow d \tan^{-1} \left(\frac{y}{x} \right) = -dx \\ \Rightarrow \int d \tan^{-1} \left(\frac{y}{x} \right) &= \int -dx \\ \therefore \tan^{-1} \left(\frac{y}{x} \right) &= -x + C \end{aligned}$$

Q31 (3) - Exact Form

$$\begin{aligned} \text{We have, } \{y(1+x^{-1}) + \sin y\} dx + \{x + \log x + x \cos y\} dy &= 0 \\ \Rightarrow ydx + \frac{y}{x}dx + \sin ydx + xdy + \log xdy + x \cos ydy &= 0 \\ \Rightarrow ydx + xdy + \frac{y}{x}dx + \log xdy + \sin ydx + x \cos ydy &= 0 \\ \Rightarrow d(xy) + d(y \log x) + d(x \sin y) &= 0 \\ \Rightarrow \int d(xy) + \int d(y \log x) + \int d(x \sin y) &= 0 \\ \therefore xy + y \log x + x \sin y &= c \end{aligned}$$

Q32 (3) - Exact Form

$$\begin{aligned} \text{We have, } x \cos \left(\frac{y}{x} \right) \cdot \{ydx + xdy\} &= y \sin \left(\frac{y}{x} \right) \{xdy - ydx\} \\ \Rightarrow xy \cos \left(\frac{y}{x} \right) dx + x^2 \cos \frac{y}{x} dy &= xy \sin \left(\frac{y}{x} \right) dy - y^2 \sin \left(\frac{y}{x} \right) dx \\ \Rightarrow x^2 \cos \frac{y}{x} dy - xy \sin \left(\frac{y}{x} \right) dy &= - \left\{ y^2 \sin \left(\frac{y}{x} \right) + xy \cos \left(\frac{y}{x} \right) \right\} dx \\ \Rightarrow \frac{dy}{dx} &= \frac{- \left[y^2 \sin \left(\frac{y}{x} \right) + xy \cos \left(\frac{y}{x} \right) \right]}{x^2 \cos \left(\frac{y}{x} \right) - xy \sin \left(\frac{y}{x} \right)} \\ \Rightarrow v + x \frac{dv}{dx} &= \frac{-(v^2 \sin v + v \cos v)}{\cos v - v \sin v} \left[\because y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx} \right] \\ \Rightarrow x \frac{dv}{dx} &= \frac{-v^2 \sin v - v \cos v - v \cos v + v^2 \sin v}{\cos v - v \sin v} \\ \Rightarrow x \frac{dv}{dx} &= \frac{-2v \cos v}{\cos v - v \sin v} \end{aligned}$$

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$$\Rightarrow \frac{\cos v - v \sin v}{v \cos v} dv = \frac{-2dx}{x}$$

$$\Rightarrow \int \left(\frac{1}{v} - \tan v \right) dV = -2 \int \frac{dx}{x}$$

$$\Rightarrow \log v - \log \sec v = -2 \log x + \log C$$

$$\Rightarrow \log v + \log x^2 = \log C + \log \sec v$$

$$\Rightarrow vx^2 = C \sec v$$

$$\Rightarrow \frac{y}{x} \cdot x^2 = C \sec \left(\frac{y}{x} \right) \left[\because v = \frac{y}{x} \right]$$

$$\therefore xy = C \sec \left(\frac{y}{x} \right)$$

Q33 (3) - Orthogonal Trajectory

We have, $y = cx^2$... (i)

On differentiating w.r.t. x , we get

$$\frac{dy}{dx} = 2cx \text{ ... (ii)}$$

From Eqs. (i) and (ii), we get

$$y = \frac{1}{2} x \frac{dy}{dx}$$

$$\Rightarrow 2y = x \frac{dy}{dx} \text{ ... (iii)}$$

Now, to obtain orthogonal trajectory replace $\frac{dy}{dx}$ by $\left(-\frac{dx}{dy} \right)$

Eq. (iii), we get

$$\Rightarrow 2y = -x \frac{dx}{dy}$$

$$\Rightarrow 2ydy = -x dx$$

On integrating both sides, we get

$$\int 2ydy = - \int x dx$$

$$\Rightarrow y^2 = -\frac{x^2}{2} + C_1$$

$$\therefore x^2 + 2y^2 = 2C_1$$

Q34 (4) - Orthogonal Trajectory

We have, $x^2 + y^2 - 2cx = 0$... (i)

On differentiating w.r.t. x both sides, we get

$$2x + 2y \frac{dy}{dx} - 2c = 0$$

On putting the value of c in Eq. (i), we get

Hints & Solutions

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$$x^2 + y^2 - 2 \left(x + y \frac{dy}{dx} \right) x = 0$$

$$\Rightarrow y^2 - x^2 = 2xy \frac{dy}{dx}$$

For orthogonal trajectories, we replace $\frac{dy}{dx}$ by $\left(-\frac{dx}{dy}\right)$ we get

$$y^2 - x^2 = -2xy \frac{dx}{dy} \Rightarrow \frac{dy}{dx} = \frac{2xy}{x^2 - y^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{2v}{1-v^2} \quad \left[\because y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx} \right]$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2v - v + v^3}{1-v^2}$$

On integrating both sides, we get

$$\int \frac{1-v^2}{v+v^3} dv = \int \frac{dx}{x}$$

$$\Rightarrow \int \left(\frac{1}{v} - \frac{2v}{1+v^2} \right) dv = \int \frac{dx}{x}$$

$$\Rightarrow \log v - \log(1+v^2) = \log x + \log C$$

$$\Rightarrow \frac{v}{1+v^2} = xC$$

$$\Rightarrow \frac{yx}{x^2+y^2} = Cx$$

$$\therefore x^2 + y^2 + Cy = 0$$